Question no .1

What is the domain and range of $y = \arcsin x$ and $\arccos x$?

Solution:

The domain of $y = \arcsin x$ is [-1,1]

Range of $y = \arcsin x$ = [-pi/2, pi/2]

Domain of $y = \arccos x$ = [-1,1]

Range of $y = \arccos x$ = [0, pi]

Question no. 2

Prove the Pythagorean Identities.

Solution:

$$\tan^2\theta + 1 = \sec^2\theta$$

$$\left(\frac{\sin\theta}{\cos\theta}\right)^2 + 1 = \frac{\sin^2\theta}{\cos^2\theta} + 1$$

$$= \frac{\sin^2\theta + \cos^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta} = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

Solution:

Use $\cot\theta = \frac{\cos\theta}{\sin\theta}$ and solve as above.

Question no.3

Expand the sum and difference formula

Sin (A+B) and Sin (A-B)

Solution:

Given in handouts.

Question no. 4

Use trigonometric identities to prove that.

$$\frac{\tan y}{\sin y} = \sec y$$

Solution:

$$\frac{\tan y}{\sin y} = \frac{\sin y}{\cos y}.\frac{1}{\sin y} = \frac{1}{\cos y} = \sec y$$

Question no .5

Without using the calculator, find $arc\sin\frac{1}{\sqrt{2}}$ in radians.

Solution:

$arc\sin\frac{1}{\sqrt{2}} = \frac{\pi}{4}$ since $\frac{\pi}{4}$ is the angle whose sine is $\frac{1}{\sqrt{2}}$.

Question no .6

Without using the calculator, find the exact value of Cos 210°.

Solution:

$$\begin{aligned}\cos(210°) &= \cos(180° + 30°)\\ &= \cos 180°\cos 30° - \sin 180°\sin 30°\\ &= (-1)\left(\frac{\sqrt{3}}{2}\right) - (0)\left(\frac{1}{2}\right)\\ &= -\frac{\sqrt{3}}{2}\end{aligned}$$

Question no .7

Given that $\sin\theta = \frac{\sqrt{3}}{2}$, find all possible values of θ in the range *-180 ≤ θ ≤ 180.*

Solution:

$\theta = \sin^{-1}(\frac{\sqrt{3}}{2}) = 60^\circ$

Also by using symmetry property:

$\because \sin(180-\theta) = \sin(\theta) \Rightarrow \sin(180-(-60)) = \sin(240)$ to get 240 as a solution.

Note that 240 must be a solution but it does not lies in our range *-180 ≤ θ ≤ 180.*

Now use the periodic property: *sin(240) = sin(240 - 360).* So another solution is -*120*degrees.

All the solutions are ± 60 and ± 120 degrees.

Question no .8

Verify that the following:

$$\frac{1-\tan\theta}{1-\cot\theta} = -\tan\theta.$$

Solution:

$$\because \frac{1-\tan\theta}{1-\cot\theta} = -\tan\theta$$

$$LHS = \frac{1-\tan\theta}{1-\cot\theta}$$

$$= \frac{1-\dfrac{\sin\theta}{\cos\theta}}{1-\dfrac{\cos\theta}{\sin\theta}}$$

$$= \frac{\dfrac{\cos\theta-\sin\theta}{\cos\theta}}{\dfrac{\sin\theta-\cos\theta}{\sin\theta}}$$

$$= \frac{\cos\theta-\sin\theta}{\cos\theta} \times \frac{\sin\theta}{-(\cos\theta-\sin\theta)}$$

$$= -\frac{\sin\theta}{\cos\theta}$$

$$\frac{1+\tan\theta}{1+\cot\theta} = -\tan\theta = RHS \quad (\text{verified})$$

Question no .9

Verify that $\cos 4A = \cos^2 2A - \sin^2 2A$.

Solution:

$$\cos 4A = \cos(2A+2A)$$

$$= \cos 2A\cos 2A - \sin 2A\sin 2A$$

$$= \cos^2 2A - \sin^2 2A \quad \text{(verified)}$$

Question no .10

Solution:

Write the double angle formula for $\tan$ function.

$$\tan 2A = \frac{\tan A+\tan A}{1-\tan A\tan A} = \frac{2\tan A}{1-\tan^2 A}$$